

CORE PURE / MISCELLANEOUS PURE

Roots of Polynomials

Worked solutions

15 questions · 117 marks

Questions

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1. The cubic equation

$$x^3 - 5x^2 + px - q = 0$$

where p and q are positive real constants, has roots α , β and γ .

Given that

$$(\alpha - \beta)^2 + (\beta - \gamma)^2 + (\gamma - \alpha)^2 = 14$$

(a) show that $p = 6$

[3]

Given that

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = 3$$

(b) determine the value of q

[3]

(c) Without solving the cubic equation, determine the value of $(\alpha + 1)(\beta + 1)(\gamma + 1)$

[4]

Solution

(a) For the cubic

$$x^3 - 5x^2 + px - q = 0$$

with roots α, β, γ , the relations between roots and coefficients give

$$\alpha + \beta + \gamma = 5, \quad \alpha\beta + \beta\gamma + \gamma\alpha = p$$

Now expand the given expression:

$$\begin{aligned} (\alpha - \beta)^2 + (\beta - \gamma)^2 + (\gamma - \alpha)^2 &= (\alpha^2 + \beta^2 - 2\alpha\beta) + (\beta^2 + \gamma^2 - 2\beta\gamma) + (\gamma^2 + \alpha^2 - 2\gamma\alpha) \\ &= 2(\alpha^2 + \beta^2 + \gamma^2) - 2(\alpha\beta + \beta\gamma + \gamma\alpha) \end{aligned}$$

Also,

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)$$

So

$$\begin{aligned} (\alpha - \beta)^2 + (\beta - \gamma)^2 + (\gamma - \alpha)^2 &= 2((\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)) - 2(\alpha\beta + \beta\gamma + \gamma\alpha) \\ &= 2((\alpha + \beta + \gamma)^2 - 3(\alpha\beta + \beta\gamma + \gamma\alpha)) \end{aligned}$$

Substituting the known values,

$$14 = 2(5^2 - 3p)$$

$$14 = 2(25 - 3p)$$

$$7 = 25 - 3p$$

$$3p = 18$$

$$p = 6$$

Hence, $p = 6$.

(b) For this cubic,

$$\alpha\beta\gamma = q$$

Also,

$$\begin{aligned} \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} &= \frac{\beta\gamma + \gamma\alpha + \alpha\beta}{\alpha\beta\gamma} \\ &= \frac{p}{q} \end{aligned}$$

Given that this sum is 3,

$$\frac{p}{q} = 3$$

From part (a), $p = 6$, so

$$\begin{aligned}\frac{6}{q} &= 3 \\ 6 &= 3q \\ q &= 2\end{aligned}$$

(c) Expand:

$$(\alpha + 1)(\beta + 1)(\gamma + 1) = \alpha\beta\gamma + (\alpha\beta + \beta\gamma + \gamma\alpha) + (\alpha + \beta + \gamma) + 1$$

Using the root relations,

$$\alpha\beta\gamma = q, \quad \alpha\beta + \beta\gamma + \gamma\alpha = p, \quad \alpha + \beta + \gamma = 5$$

Therefore

$$\begin{aligned}(\alpha + 1)(\beta + 1)(\gamma + 1) &= q + p + 5 + 1 \\ &= 2 + 6 + 5 + 1 \\ &= 14\end{aligned}$$

2. The quartic equation $x^4 + bx^3 + cx^2 + dx + 3 = 0$ has roots $\alpha, \beta, \gamma, \delta$. It is given that

$$\alpha + \beta + \gamma + \delta = -2, \quad (\alpha + 1)^2 + (\beta + 1)^2 + (\gamma + 1)^2 + (\delta + 1)^2 = 6, \quad \alpha^{-1} + \beta^{-1} + \gamma^{-1} + \delta^{-1} = 2$$

(a) Find the values of b, c and d [6]

(b) Given also that

$$(\alpha + 1)^3 + (\beta + 1)^3 + (\gamma + 1)^3 + (\delta + 1)^3 = 20$$

find the value of $\alpha^4 + \beta^4 + \gamma^4 + \delta^4$ [2]

Solution

(a) For the monic quartic

$$x^4 + bx^3 + cx^2 + dx + 3 = 0$$

with roots $\alpha, \beta, \gamma, \delta$, the standard relations between coefficients and roots give

$$\alpha + \beta + \gamma + \delta = -b$$

$$\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = c$$

$$\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta = -d$$

$$\alpha\beta\gamma\delta = 3$$

We are given

$$\alpha + \beta + \gamma + \delta = -2$$

so

$$-b = -2$$

hence

$$b = 2$$

Next,

$$\alpha^{-1} + \beta^{-1} + \gamma^{-1} + \delta^{-1} = \frac{\beta\gamma\delta + \alpha\gamma\delta + \alpha\beta\delta + \alpha\beta\gamma}{\alpha\beta\gamma\delta}$$

Therefore

$$2 = \frac{-d}{3}$$

so

$$d = -6$$

Now use

$$(\alpha + 1)^2 + (\beta + 1)^2 + (\gamma + 1)^2 + (\delta + 1)^2 = 6$$

Expanding,

$$\alpha^2 + \beta^2 + \gamma^2 + \delta^2 + 2(\alpha + \beta + \gamma + \delta) + 4 = 6$$

since there are 4 lots of 1^2 .

Substituting $\alpha + \beta + \gamma + \delta = -2$,

$$\alpha^2 + \beta^2 + \gamma^2 + \delta^2 + 2(-2) + 4 = 6$$

$$\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = 6$$

Also,

$$(\alpha + \beta + \gamma + \delta)^2 = \alpha^2 + \beta^2 + \gamma^2 + \delta^2 + 2(\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta)$$

so

$$(-2)^2 = 6 + 2c$$

$$4 = 6 + 2c$$

$$2c = -2$$

$$c = -1$$

Therefore,

$$b = 2, \quad c = -1, \quad d = -6$$

(b) Using the extra condition

$$(\alpha + 1)^3 + (\beta + 1)^3 + (\gamma + 1)^3 + (\delta + 1)^3 = 20$$

expand:

$$\sum (\text{root} + 1)^3 = \sum \text{root}^3 + 3 \sum \text{root}^2 + 3 \sum \text{root} + 4$$

Hence

$$\alpha^3 + \beta^3 + \gamma^3 + \delta^3 + 3(\alpha^2 + \beta^2 + \gamma^2 + \delta^2) + 3(\alpha + \beta + \gamma + \delta) + 4 = 20$$

Using $\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = 6$ and $\alpha + \beta + \gamma + \delta = -2$,

$$\alpha^3 + \beta^3 + \gamma^3 + \delta^3 + 3(6) + 3(-2) + 4 = 20$$

$$\alpha^3 + \beta^3 + \gamma^3 + \delta^3 + 18 - 6 + 4 = 20$$

$$\alpha^3 + \beta^3 + \gamma^3 + \delta^3 = 4$$

From part (a), the quartic is

$$x^4 + 2x^3 - x^2 - 6x + 3 = 0$$

so each root r satisfies

$$r^4 + 2r^3 - r^2 - 6r + 3 = 0$$

and therefore

$$r^4 = -2r^3 + r^2 + 6r - 3$$

Summing this for $r = \alpha, \beta, \gamma, \delta$ gives

$$\begin{aligned} \alpha^4 + \beta^4 + \gamma^4 + \delta^4 &= -2(\alpha^3 + \beta^3 + \gamma^3 + \delta^3) + (\alpha^2 + \beta^2 + \gamma^2 + \delta^2) + 6(\alpha + \beta + \gamma + \delta) - 12 \\ &= -2(4) + 6 + 6(-2) - 12 \\ &= -8 + 6 - 12 - 12 \\ &= -26 \end{aligned}$$

Therefore,

$$\alpha^4 + \beta^4 + \gamma^4 + \delta^4 = -26$$

3. The equation $x^3 - 4x^2 - x + 6 = 0$ has roots α , β and γ .

Determine the value of

$$\frac{1}{\alpha + \beta} + \frac{1}{\beta + \gamma} + \frac{1}{\gamma + \alpha}$$

[4]

Solution

Let

$$s_1 = \alpha + \beta + \gamma, \quad s_2 = \alpha\beta + \beta\gamma + \gamma\alpha, \quad s_3 = \alpha\beta\gamma$$

From

$$x^3 - 4x^2 - x + 6 = 0$$

comparing with $x^3 - s_1x^2 + s_2x - s_3 = 0$, we get

$$s_1 = 4, \quad s_2 = -1, \quad s_3 = -6$$

We need

$$\frac{1}{\alpha + \beta} + \frac{1}{\beta + \gamma} + \frac{1}{\gamma + \alpha}$$

Taking a common denominator,

$$\frac{1}{\alpha + \beta} + \frac{1}{\beta + \gamma} + \frac{1}{\gamma + \alpha} = \frac{(\beta + \gamma)(\gamma + \alpha) + (\gamma + \alpha)(\alpha + \beta) + (\alpha + \beta)(\beta + \gamma)}{(\alpha + \beta)(\beta + \gamma)(\gamma + \alpha)}$$

Expand the numerator. Each product, such as $(\alpha + \beta)(\beta + \gamma) = \beta^2 + \alpha\beta + \beta\gamma + \gamma\alpha$, gives one square and all three pairwise products, so

$$\begin{aligned} (\beta + \gamma)(\gamma + \alpha) + (\gamma + \alpha)(\alpha + \beta) + (\alpha + \beta)(\beta + \gamma) &= (\alpha^2 + \beta^2 + \gamma^2) + 3(\alpha\beta + \beta\gamma + \gamma\alpha) \\ &= (s_1^2 - 2s_2) + 3s_2 \\ &= s_1^2 + s_2 \end{aligned}$$

Now expand the denominator:

$$\begin{aligned} (\alpha + \beta)(\beta + \gamma)(\gamma + \alpha) &= (\alpha + \beta)(\beta\gamma + \beta\alpha + \gamma^2 + \gamma\alpha) \\ &= \alpha^2\beta + \alpha^2\gamma + \alpha\beta^2 + \beta^2\gamma + \alpha\gamma^2 + \beta\gamma^2 + 2\alpha\beta\gamma \end{aligned}$$

Also,

$$(\alpha + \beta + \gamma)(\alpha\beta + \beta\gamma + \gamma\alpha) = \alpha^2\beta + \alpha^2\gamma + \alpha\beta^2 + \beta^2\gamma + \alpha\gamma^2 + \beta\gamma^2 + 3\alpha\beta\gamma$$

So

$$(\alpha + \beta)(\beta + \gamma)(\gamma + \alpha) = s_1s_2 - s_3$$

Hence

$$\frac{1}{\alpha + \beta} + \frac{1}{\beta + \gamma} + \frac{1}{\gamma + \alpha} = \frac{s_1^2 + s_2}{s_1s_2 - s_3}$$

Substitute $s_1 = 4$, $s_2 = -1$, $s_3 = -6$:

$$\begin{aligned} \frac{s_1^2 + s_2}{s_1s_2 - s_3} &= \frac{4^2 + (-1)}{4(-1) - (-6)} \\ &= \frac{16 - 1}{-4 + 6} \\ &= \frac{15}{2} \end{aligned}$$

Therefore, the required value is $\frac{15}{2}$.

4. The quartic equation $3x^4 - x^3 - 10x^2 + 8x + 4 = 0$ has roots α, β, γ and δ .

Without solving the equation, find equations with integer coefficients whose roots are

(a) $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$ and $\frac{1}{\delta}$ [6]

(b) $\frac{2}{\alpha-1}, \frac{2}{\beta-1}, \frac{2}{\gamma-1}$ and $\frac{2}{\delta-1}$ [6]

Solution

(a) Let

$$f(x) = 3x^4 - x^3 - 10x^2 + 8x + 4$$

and let $x = \alpha$ be any root, so $f(\alpha) = 0$.

We want an equation whose roots are

$$\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}, \frac{1}{\delta}$$

Since the constant term of $f(x)$ is 4, none of the roots is 0, so reciprocals are defined.

Let

$$y = \frac{1}{x}$$

so that

$$x = \frac{1}{y}$$

Substitute $x = \frac{1}{y}$ into $f(x) = 0$:

$$3\left(\frac{1}{y}\right)^4 - \left(\frac{1}{y}\right)^3 - 10\left(\frac{1}{y}\right)^2 + 8\left(\frac{1}{y}\right) + 4 = 0$$

This gives

$$\frac{3}{y^4} - \frac{1}{y^3} - \frac{10}{y^2} + \frac{8}{y} + 4 = 0$$

Multiply through by y^4 :

$$3 - y - 10y^2 + 8y^3 + 4y^4 = 0$$

Rearrange in descending powers:

$$4y^4 + 8y^3 - 10y^2 - y + 3 = 0$$

So the equation whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}, \frac{1}{\delta}$ is

$$4x^4 + 8x^3 - 10x^2 - x + 3 = 0$$

(b) We now want an equation whose roots are

$$\frac{2}{\alpha-1}, \frac{2}{\beta-1}, \frac{2}{\gamma-1}, \frac{2}{\delta-1}$$

First check that 1 is not a root of the original equation:

$$f(1) = 3 - 1 - 10 + 8 + 4 = 4 \neq 0$$

so $\alpha - 1, \beta - 1, \gamma - 1, \delta - 1$ are all non-zero.

Let

$$y = \frac{2}{x-1}$$

Then

$$x-1 = \frac{2}{y} \quad \Rightarrow \quad x = 1 + \frac{2}{y}$$

Substitute $x = 1 + \frac{2}{y}$ into $f(x) = 0$:

$$3\left(1 + \frac{2}{y}\right)^4 - \left(1 + \frac{2}{y}\right)^3 - 10\left(1 + \frac{2}{y}\right)^2 + 8\left(1 + \frac{2}{y}\right) + 4 = 0$$

Now expand each power:

$$\begin{aligned}\left(1 + \frac{2}{y}\right)^2 &= 1 + \frac{4}{y} + \frac{4}{y^2} \\ \left(1 + \frac{2}{y}\right)^3 &= 1 + \frac{6}{y} + \frac{12}{y^2} + \frac{8}{y^3} \\ \left(1 + \frac{2}{y}\right)^4 &= 1 + \frac{8}{y} + \frac{24}{y^2} + \frac{32}{y^3} + \frac{16}{y^4}\end{aligned}$$

So

$$\begin{aligned}0 &= 3\left(1 + \frac{8}{y} + \frac{24}{y^2} + \frac{32}{y^3} + \frac{16}{y^4}\right) - \left(1 + \frac{6}{y} + \frac{12}{y^2} + \frac{8}{y^3}\right) \\ &\quad - 10\left(1 + \frac{4}{y} + \frac{4}{y^2}\right) + 8\left(1 + \frac{2}{y}\right) + 4\end{aligned}$$

Expand fully:

$$\begin{aligned}0 &= \left(3 + \frac{24}{y} + \frac{72}{y^2} + \frac{96}{y^3} + \frac{48}{y^4}\right) - \left(1 + \frac{6}{y} + \frac{12}{y^2} + \frac{8}{y^3}\right) \\ &\quad - \left(10 + \frac{40}{y} + \frac{40}{y^2}\right) + \left(8 + \frac{16}{y}\right) + 4\end{aligned}$$

Collect like terms:

$$\begin{aligned}0 &= (3 - 1 - 10 + 8 + 4) + \left(\frac{24}{y} - \frac{6}{y} - \frac{40}{y} + \frac{16}{y}\right) \\ &\quad + \left(\frac{72}{y^2} - \frac{12}{y^2} - \frac{40}{y^2}\right) + \left(\frac{96}{y^3} - \frac{8}{y^3}\right) + \frac{48}{y^4}\end{aligned}$$

Hence

$$0 = 4 - \frac{6}{y} + \frac{20}{y^2} + \frac{88}{y^3} + \frac{48}{y^4}$$

Multiply through by y^4 :

$$4y^4 - 6y^3 + 20y^2 + 88y + 48 = 0$$

All coefficients are even, so divide by 2:

$$2y^4 - 3y^3 + 10y^2 + 44y + 24 = 0$$

Therefore the equation whose roots are $\frac{2}{\alpha-1}, \frac{2}{\beta-1}, \frac{2}{\gamma-1}, \frac{2}{\delta-1}$ is

$$2x^4 - 3x^3 + 10x^2 + 44x + 24 = 0$$

5. The cubic equation $x^3 + x^2 - 2x - 1 = 0$ has roots α , β and γ .

(a) Find a cubic equation whose roots are $\frac{1}{\alpha+1}$, $\frac{1}{\beta+1}$ and $\frac{1}{\gamma+1}$ [3]

(b) Find the value of

$$\left(\frac{1}{\alpha+1}\right)^2 + \left(\frac{1}{\beta+1}\right)^2 + \left(\frac{1}{\gamma+1}\right)^2 \quad [2]$$

(c) Find the value of

$$\left(\frac{1}{\alpha+1}\right)^3 + \left(\frac{1}{\beta+1}\right)^3 + \left(\frac{1}{\gamma+1}\right)^3 \quad [2]$$

Solution

(a) Since $x = -1$ is not a root of $x^3 + x^2 - 2x - 1 = 0$, the quantities

$$\frac{1}{\alpha+1}, \quad \frac{1}{\beta+1}, \quad \frac{1}{\gamma+1}$$

are all defined.

Let

$$y = \frac{1}{x+1}$$

so

$$x = \frac{1}{y} - 1 = \frac{1-y}{y}$$

Substitute into the given cubic:

$$\left(\frac{1-y}{y}\right)^3 + \left(\frac{1-y}{y}\right)^2 - 2\left(\frac{1-y}{y}\right) - 1 = 0$$

Multiply through by y^3 :

$$(1-y)^3 + y(1-y)^2 - 2y^2(1-y) - y^3 = 0$$

Expanding:

$$(1 - 3y + 3y^2 - y^3) + (y - 2y^2 + y^3) - 2y^2 + 2y^3 - y^3 = 0$$

Collecting terms:

$$1 - 2y - y^2 + y^3 = 0$$

So

$$y^3 - y^2 - 2y + 1 = 0$$

When $x = \alpha, \beta, \gamma$, the corresponding values of y are

$$\frac{1}{\alpha+1}, \quad \frac{1}{\beta+1}, \quad \frac{1}{\gamma+1}$$

so the required cubic equation is

$$y^3 - y^2 - 2y + 1 = 0$$

(b) Let the roots of

$$y^3 - y^2 - 2y + 1 = 0$$

be p, q, r , where

$$p = \frac{1}{\alpha + 1}, \quad q = \frac{1}{\beta + 1}, \quad r = \frac{1}{\gamma + 1}$$

From the cubic,

$$p + q + r = 1, \quad pq + pr + qr = -2$$

Now

$$\begin{aligned} p^2 + q^2 + r^2 &= (p + q + r)^2 - 2(pq + pr + qr) \\ &= 1^2 - 2(-2) \\ &= 5 \end{aligned}$$

Therefore

$$\left(\frac{1}{\alpha + 1}\right)^2 + \left(\frac{1}{\beta + 1}\right)^2 + \left(\frac{1}{\gamma + 1}\right)^2 = 5$$

(c) Using the same roots p, q, r , we also have

$$pqr = -1$$

Use

$$p^3 + q^3 + r^3 = (p + q + r)^3 - 3(p + q + r)(pq + pr + qr) + 3pqr$$

Hence

$$\begin{aligned} p^3 + q^3 + r^3 &= 1^3 - 3(1)(-2) + 3(-1) \\ &= 1 + 6 - 3 \\ &= 4 \end{aligned}$$

Therefore

$$\left(\frac{1}{\alpha + 1}\right)^3 + \left(\frac{1}{\beta + 1}\right)^3 + \left(\frac{1}{\gamma + 1}\right)^3 = 4$$

6. The equation $x^3 - 14x^2 + 56x + c = 0$, where c is a constant, has three roots in geometric progression.

(a) Determine the roots of the equation [6]

(b) Find c [1]

Solution

(a) Let the three roots be

$$\frac{a}{r}, \quad a, \quad ar$$

where $r \neq 0$.

For the cubic

$$x^3 - 14x^2 + 56x + c = 0$$

the sum of the roots is 14, and the sum of the products of the roots taken two at a time is 56.

So

$$\begin{aligned} \frac{a}{r} + a + ar &= 14 \\ a \left(\frac{1}{r} + 1 + r \right) &= 14 \end{aligned}$$

Also,

$$\begin{aligned} \left(\frac{a}{r} \right) (a) + a(ar) + \left(\frac{a}{r} \right) (ar) &= 56 \\ \frac{a^2}{r} + a^2r + a^2 &= 56 \\ a^2 \left(\frac{1}{r} + 1 + r \right) &= 56 \end{aligned}$$

Now divide the second equation by the first:

$$\frac{a^2 \left(\frac{1}{r} + 1 + r \right)}{a \left(\frac{1}{r} + 1 + r \right)} = \frac{56}{14}$$

so

$$a = 4$$

Substitute into

$$a \left(\frac{1}{r} + 1 + r \right) = 14$$

to get

$$\begin{aligned} 4 \left(\frac{1}{r} + 1 + r \right) &= 14 \\ \frac{1}{r} + 1 + r &= \frac{7}{2} \end{aligned}$$

Multiply through by $2r$:

$$\begin{aligned} 2 + 2r + 2r^2 &= 7r \\ 2r^2 - 5r + 2 &= 0 \end{aligned}$$

Factorising,

$$(2r - 1)(r - 2) = 0$$

so

$$r = \frac{1}{2} \quad \text{or} \quad r = 2$$

Both values of r give the same three roots: $r = 2$ gives

$$\frac{4}{2}, 4, 4(2)$$

and $r = \frac{1}{2}$ gives them in the reverse order. So the roots are

$$2, 4, 8$$

(b) The product of the roots is

$$2 \cdot 4 \cdot 8 = 64$$

For

$$x^3 - 14x^2 + 56x + c = 0$$

the product of the roots is $-c$.

So

$$-c = 64$$

and therefore

$$c = -64$$

7. The quadratic equation

$$x^2 - kx + 1 = 0$$

where k is an integer, has roots α and β .

(a) Write down, in terms of k where appropriate, the value of $\alpha + \beta$ and the value of $\alpha\beta$ [2]

(b) Determine, in simplest form in terms of k , the value of

$$(\alpha^2 + \beta) + (\beta^2 + \alpha) \quad [3]$$

(c) Determine a quadratic equation which has roots

$$\alpha^2 + \beta \text{ and } \beta^2 + \alpha$$

giving your answer in the form $px^2 + qx + r = 0$, where p , q and r are integer values in terms of k [4]

Solution

(a) For the quadratic $x^2 - kx + 1 = 0$, we use

$$\alpha + \beta = -\frac{\text{coefficient of } x}{\text{coefficient of } x^2}, \quad \alpha\beta = \frac{\text{constant term}}{\text{coefficient of } x^2}$$

So

$$\alpha + \beta = -\frac{-k}{1} = k, \quad \alpha\beta = \frac{1}{1} = 1$$

(b)

$$\begin{aligned} (\alpha^2 + \beta) + (\beta^2 + \alpha) &= \alpha^2 + \beta^2 + \alpha + \beta \\ &= (\alpha + \beta)^2 - 2\alpha\beta + (\alpha + \beta) \\ &= k^2 - 2(1) + k \\ &= k^2 + k - 2 \end{aligned}$$

(c) Let the new roots be

$$\alpha^2 + \beta \quad \text{and} \quad \beta^2 + \alpha$$

Their sum is, from part (b),

$$(\alpha^2 + \beta) + (\beta^2 + \alpha) = k^2 + k - 2$$

Now find their product:

$$\begin{aligned} (\alpha^2 + \beta)(\beta^2 + \alpha) &= \alpha^2\beta^2 + \alpha^3 + \beta^3 + \alpha\beta \\ &= (\alpha\beta)^2 + \alpha^3 + \beta^3 + \alpha\beta \\ &= 1 + \alpha^3 + \beta^3 + 1 \\ &= \alpha^3 + \beta^3 + 2 \end{aligned}$$

Next,

$$\begin{aligned} \alpha^3 + \beta^3 &= (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) \\ &= k^3 - 3(1)(k) \\ &= k^3 - 3k \end{aligned}$$

So the product is

$$(\alpha^2 + \beta)(\beta^2 + \alpha) = k^3 - 3k + 2$$

A quadratic with roots r_1 and r_2 is

$$x^2 - (r_1 + r_2)x + r_1 r_2 = 0$$

Therefore the required equation is

$$x^2 - (k^2 + k - 2)x + (k^3 - 3k + 2) = 0$$

which is in the form $px^2 + qx + r = 0$ with $p = 1$, $q = -(k^2 + k - 2)$, $r = k^3 - 3k + 2$.

8. The roots of the equation

$$x^3 + 5x^2 - 4x - 24 = 0$$

are α , β and γ .

Without solving the equation,

(a) write down the value of each of

$$\alpha + \beta + \gamma \quad \alpha\beta + \beta\gamma + \gamma\alpha \quad \alpha\beta\gamma \quad [1]$$

(b) Hence determine the value of

(i)

$$\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\gamma\alpha} \quad [2]$$

(ii)

$$(\alpha + 2)(\beta + 2)(\gamma + 2) \quad [2]$$

(iii)

$$\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 \quad [3]$$

Solution

(a) For a monic cubic

$$x^3 + bx^2 + cx + d = 0$$

with roots α, β, γ ,

$$\alpha + \beta + \gamma = -b, \quad \alpha\beta + \beta\gamma + \gamma\alpha = c, \quad \alpha\beta\gamma = -d$$

Here

$$x^3 + 5x^2 - 4x - 24 = 0$$

so

$$\alpha + \beta + \gamma = -5, \quad \alpha\beta + \beta\gamma + \gamma\alpha = -4, \quad \alpha\beta\gamma = 24$$

Therefore the values are -5 , -4 and 24 .

(b) Using

$$\alpha + \beta + \gamma = -5, \quad \alpha\beta + \beta\gamma + \gamma\alpha = -4, \quad \alpha\beta\gamma = 24$$

(i)

$$\begin{aligned} \frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\gamma\alpha} &= \frac{\gamma}{\alpha\beta\gamma} + \frac{\alpha}{\alpha\beta\gamma} + \frac{\beta}{\alpha\beta\gamma} \\ &= \frac{\alpha + \beta + \gamma}{\alpha\beta\gamma} \\ &= -\frac{5}{24} \end{aligned}$$

(ii)

$$\begin{aligned} (\alpha + 2)(\beta + 2)(\gamma + 2) &= (\alpha\beta + 2\alpha + 2\beta + 4)(\gamma + 2) \\ &= \alpha\beta\gamma + 2\alpha\beta + 2\alpha\gamma + 2\beta\gamma + 4\alpha + 4\beta + 4\gamma + 8 \\ &= \alpha\beta\gamma + 2(\alpha\beta + \beta\gamma + \gamma\alpha) + 4(\alpha + \beta + \gamma) + 8 \end{aligned}$$

Substituting the values from part (a),

$$\begin{aligned} (\alpha + 2)(\beta + 2)(\gamma + 2) &= 24 + 2(-4) + 4(-5) + 8 \\ &= 24 - 8 - 20 + 8 \\ &= 4 \end{aligned}$$

(iii) Start with

$$(\alpha\beta + \beta\gamma + \gamma\alpha)^2$$

Expanding,

$$\begin{aligned}(\alpha\beta + \beta\gamma + \gamma\alpha)^2 &= \alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 + 2\alpha\beta \cdot \beta\gamma + 2\beta\gamma \cdot \gamma\alpha + 2\gamma\alpha \cdot \alpha\beta \\&= \alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 + 2\alpha\beta^2\gamma + 2\alpha\beta\gamma^2 + 2\alpha^2\beta\gamma \\&= \alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 + 2\alpha\beta\gamma(\alpha + \beta + \gamma)\end{aligned}$$

So

$$\begin{aligned}\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 &= (\alpha\beta + \beta\gamma + \gamma\alpha)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma) \\&= (-4)^2 - 2(24)(-5) \\&= 16 + 240 \\&= 256\end{aligned}$$

9. The roots of the equation $3x^3 + px^2 - 12x - 8 = 0$ are α , β and γ .

(a) Given that $\alpha + \beta + \gamma = 2$, write down the value of p [1]

(b) Write down values for $\alpha\beta + \beta\gamma + \gamma\alpha$ and $\alpha\beta\gamma$ [1]

(c) Hence find the value of $\left(1 + \frac{1}{\alpha}\right)\left(1 + \frac{1}{\beta}\right)\left(1 + \frac{1}{\gamma}\right)$ [3]

Solution

(a) For $3x^3 + px^2 - 12x - 8 = 0$, the sum of the roots is

$$\alpha + \beta + \gamma = -\frac{p}{3}$$

Given $\alpha + \beta + \gamma = 2$,

$$-\frac{p}{3} = 2$$

so

$$p = -6$$

(b) For $3x^3 + px^2 - 12x - 8 = 0$,

$$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{-12}{3} = -4$$

and

$$\alpha\beta\gamma = -\frac{-8}{3} = \frac{8}{3}$$

Therefore,

$$\alpha\beta + \beta\gamma + \gamma\alpha = -4, \quad \alpha\beta\gamma = \frac{8}{3}$$

(c)

$$\begin{aligned} \left(1 + \frac{1}{\alpha}\right)\left(1 + \frac{1}{\beta}\right)\left(1 + \frac{1}{\gamma}\right) &= \frac{(\alpha+1)(\beta+1)(\gamma+1)}{\alpha\beta\gamma} \\ &= \frac{\alpha\beta\gamma + \alpha\beta + \beta\gamma + \gamma\alpha + \alpha + \beta + \gamma + 1}{\alpha\beta\gamma} \end{aligned}$$

Using the results from parts (a) and (b),

$$\alpha + \beta + \gamma = 2, \quad \alpha\beta + \beta\gamma + \gamma\alpha = -4, \quad \alpha\beta\gamma = \frac{8}{3}$$

so

$$\begin{aligned} \left(1 + \frac{1}{\alpha}\right)\left(1 + \frac{1}{\beta}\right)\left(1 + \frac{1}{\gamma}\right) &= \frac{\frac{8}{3} - 4 + 2 + 1}{\frac{8}{3}} \\ &= \frac{\frac{8}{3} - 1}{\frac{8}{3}} \\ &= \frac{\frac{5}{3}}{\frac{8}{3}} \\ &= \frac{5}{8} \end{aligned}$$

10. The roots of the equation $x^3 + 2x^2 - 5x - 3 = 0$ are α , β and γ .

(a) Find $\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2$ [4]

(b) Find an equation with integer coefficients whose roots are α^2 , β^2 and γ^2 [4]

Solution

(a) Since α , β and γ are roots of

$$x^3 + 2x^2 - 5x - 3 = 0$$

the standard relations between roots and coefficients give

$$\alpha + \beta + \gamma = -2, \quad \alpha\beta + \beta\gamma + \gamma\alpha = -5, \quad \alpha\beta\gamma = 3$$

Now expand

$$\begin{aligned} (\alpha\beta + \beta\gamma + \gamma\alpha)^2 &= \alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 + 2(\alpha\beta)(\beta\gamma) + 2(\beta\gamma)(\gamma\alpha) + 2(\gamma\alpha)(\alpha\beta) \\ &= \alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 + 2\alpha\beta^2\gamma + 2\beta\gamma^2\alpha + 2\gamma\alpha^2\beta \\ &= \alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 + 2\alpha\beta\gamma(\alpha + \beta + \gamma) \end{aligned}$$

So

$$\begin{aligned} \alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 &= (\alpha\beta + \beta\gamma + \gamma\alpha)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma) \\ &= (-5)^2 - 2(3)(-2) \\ &= 25 + 12 \\ &= 37 \end{aligned}$$

(b) Let the required equation have roots α^2 , β^2 and γ^2 .

First find their sum:

$$\begin{aligned} \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) \\ &= (-2)^2 - 2(-5) \\ &= 4 + 10 \\ &= 14 \end{aligned}$$

From part (a),

$$\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 = 37$$

Also,

$$\alpha^2\beta^2\gamma^2 = (\alpha\beta\gamma)^2 = 3^2 = 9$$

So for roots $\alpha^2, \beta^2, \gamma^2$, the monic cubic is

$$x^3 - (\text{sum of roots})x^2 + (\text{sum of pairwise products})x - (\text{product of roots}) = 0$$

Hence the required equation is

$$x^3 - 14x^2 + 37x - 9 = 0$$

11. The graph of $y = x^3 + 3x^2 + px - 20$, where p is a constant, crosses the x -axis at three points whose x -coordinates are equally spaced.

Find the roots of

$$x^3 + 3x^2 + px - 20 = 0$$

[6]

Solution

Since the three roots are equally spaced, let them be

$$a - d, \quad a, \quad a + d$$

For

$$x^3 + 3x^2 + px - 20 = 0$$

the sum of the roots is -3 , so

$$(a - d) + a + (a + d) = -3$$

$$3a = -3$$

$$a = -1$$

Also, the product of the roots is 20 , because for a monic cubic $x^3 + \dots - 20$, the product of the roots is $-(-20) = 20$.

So

$$(a - d)a(a + d) = 20$$

Now

$$(a - d)(a + d) = a^2 - d^2$$

so

$$a(a^2 - d^2) = 20$$

Substitute $a = -1$:

$$(-1)(1 - d^2) = 20$$

$$-1 + d^2 = 20$$

$$d^2 = 21$$

$$d = \sqrt{21}$$

Therefore the roots are

$$a - d = -1 - \sqrt{21}, \quad a = -1, \quad a + d = -1 + \sqrt{21}$$

Hence the roots of the equation are

$$-1 - \sqrt{21}, \quad -1, \quad -1 + \sqrt{21}$$

(Checking: the pairwise sum is

$$(-1 - \sqrt{21})(-1) + (-1)(-1 + \sqrt{21}) + (-1 - \sqrt{21})(-1 + \sqrt{21}) = -18$$

so $p = -18$, which is consistent.)

12. The cubic equation $x^3 - 3x + 1 = 0$ has roots α , β and γ .

(a) Find a cubic equation whose roots are $\frac{1}{\alpha-1}$, $\frac{1}{\beta-1}$, $\frac{1}{\gamma-1}$. [3]

(b) Hence find the value of $\frac{1}{(\alpha-1)^2} + \frac{1}{(\beta-1)^2} + \frac{1}{(\gamma-1)^2}$. [2]

(c) Find also the value of $\frac{1}{(\alpha-1)^5} + \frac{1}{(\beta-1)^5} + \frac{1}{(\gamma-1)^5}$. [3]

Solution

(a) Let

$$t = \frac{1}{x-1}$$

so that

$$x = 1 + \frac{1}{t}$$

We substitute this into the given equation

$$x^3 - 3x + 1 = 0$$

Then

$$\begin{aligned} \left(1 + \frac{1}{t}\right)^3 - 3\left(1 + \frac{1}{t}\right) + 1 &= 0 \\ \left(1 + \frac{3}{t} + \frac{3}{t^2} + \frac{1}{t^3}\right) - 3 - \frac{3}{t} + 1 &= 0 \\ \frac{1}{t^3} + \frac{3}{t^2} - 1 &= 0 \end{aligned}$$

Multiply through by t^3 :

$$\begin{aligned} 1 + 3t - t^3 &= 0 \\ t^3 - 3t - 1 &= 0 \end{aligned}$$

So the cubic equation whose roots are

$$\frac{1}{\alpha-1}, \quad \frac{1}{\beta-1}, \quad \frac{1}{\gamma-1}$$

is

$$t^3 - 3t - 1 = 0$$

(b) Let the roots of

$$t^3 - 3t - 1 = 0$$

be

$$r_1 = \frac{1}{\alpha-1}, \quad r_2 = \frac{1}{\beta-1}, \quad r_3 = \frac{1}{\gamma-1}$$

For the cubic

$$t^3 + 0t^2 - 3t - 1 = 0$$

we have

$$r_1 + r_2 + r_3 = 0, \quad r_1r_2 + r_2r_3 + r_3r_1 = -3$$

Now

$$(r_1 + r_2 + r_3)^2 = r_1^2 + r_2^2 + r_3^2 + 2(r_1r_2 + r_2r_3 + r_3r_1)$$

So

$$\begin{aligned}r_1^2 + r_2^2 + r_3^2 &= (r_1 + r_2 + r_3)^2 - 2(r_1r_2 + r_2r_3 + r_3r_1) \\&= 0^2 - 2(-3) \\&= 6\end{aligned}$$

Hence

$$\frac{1}{(\alpha - 1)^2} + \frac{1}{(\beta - 1)^2} + \frac{1}{(\gamma - 1)^2} = 6$$

(c) Using the same roots r_1, r_2, r_3 , each root satisfies

$$r^3 - 3r - 1 = 0$$

so

$$r^3 = 3r + 1$$

Multiply by r^2 :

$$r^5 = r^2r^3 = r^2(3r + 1) = 3r^3 + r^2$$

Now substitute again for r^3 :

$$\begin{aligned}r^5 &= 3(3r + 1) + r^2 \\&= 9r + 3 + r^2\end{aligned}$$

Summing for r_1, r_2, r_3 gives

$$\begin{aligned}r_1^5 + r_2^5 + r_3^5 &= 9(r_1 + r_2 + r_3) + 3 \cdot 3 + (r_1^2 + r_2^2 + r_3^2) \\&= 9(0) + 9 + 6 \\&= 15\end{aligned}$$

Therefore

$$\frac{1}{(\alpha - 1)^5} + \frac{1}{(\beta - 1)^5} + \frac{1}{(\gamma - 1)^5} = 15$$

13. The quartic equation $x^4 - 2x^2 + 2x + 1 = 0$ has roots $\alpha, \beta, \gamma, \delta$.

(a) Find a quartic equation whose roots are $\alpha^2, \beta^2, \gamma^2, \delta^2$ and state the value of $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$. [5]

(b) Find the value of $\alpha^6 + \beta^6 + \gamma^6 + \delta^6$. [3]

(c) Find the value of $\alpha^8 + \beta^8 + \gamma^8 + \delta^8$. [2]

Solution

(a) Let

$$f(x) = x^4 - 2x^2 + 2x + 1$$

If $f(\alpha) = 0$, then $f(\alpha)f(-\alpha) = 0$. So α^2 is a root of the equation obtained from $f(x)f(-x) = 0$ after writing everything in terms of x^2 . The same will then be true for $\beta^2, \gamma^2, \delta^2$.

Now

$$f(x) = (x^2 - 1)^2 + 2x, \quad f(-x) = (x^2 - 1)^2 - 2x$$

So

$$\begin{aligned} f(x)f(-x) &= ((x^2 - 1)^2 + 2x)((x^2 - 1)^2 - 2x) \\ &= (x^2 - 1)^4 - 4x^2 \end{aligned}$$

Let $y = x^2$. Then

$$\begin{aligned} (x^2 - 1)^4 - 4x^2 &= (y - 1)^4 - 4y \\ &= y^4 - 4y^3 + 6y^2 - 4y + 1 - 4y \\ &= y^4 - 4y^3 + 6y^2 - 8y + 1 \end{aligned}$$

Hence the required quartic equation is

$$y^4 - 4y^3 + 6y^2 - 8y + 1 = 0$$

Its roots are $\alpha^2, \beta^2, \gamma^2, \delta^2$.

If these roots are y_1, y_2, y_3, y_4 , then for a monic quartic,

$$y_1 + y_2 + y_3 + y_4 = -(\text{coefficient of } y^3) = 4$$

Therefore

$$\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = 4$$

(b) Let the roots of

$$h(y) = y^4 - 4y^3 + 6y^2 - 8y + 1$$

be y_1, y_2, y_3, y_4 , where

$$y_1 = \alpha^2, \quad y_2 = \beta^2, \quad y_3 = \gamma^2, \quad y_4 = \delta^2$$

Define

$$q_r = y_1^r + y_2^r + y_3^r + y_4^r$$

Then

$$\alpha^6 + \beta^6 + \gamma^6 + \delta^6 = q_3$$

From the coefficients of $h(y)$,

$$\sum y_i = 4, \quad \sum_{i < j} y_i y_j = 6, \quad \sum_{i < j < k} y_i y_j y_k = 8$$

So $q_1 = 4$, and

$$\begin{aligned} q_2 &= \left(\sum y_i \right)^2 - 2 \sum_{i < j} y_i y_j \\ &= 4^2 - 2(6) \\ &= 4 \end{aligned}$$

The identity for a sum of cubes used with cubics holds for any number of roots, so

$$\begin{aligned} q_3 &= \left(\sum y_i \right)^3 - 3 \left(\sum y_i \right) \left(\sum_{i < j} y_i y_j \right) + 3 \sum_{i < j < k} y_i y_j y_k \\ &= 4^3 - 3(4)(6) + 3(8) \\ &= 64 - 72 + 24 \\ &= 16 \end{aligned}$$

Therefore

$$\alpha^6 + \beta^6 + \gamma^6 + \delta^6 = 16$$

(c) Here

$$\alpha^8 + \beta^8 + \gamma^8 + \delta^8 = q_4$$

Each y_i is a root of $h(y) = 0$, so

$$y_i^4 = 4y_i^3 - 6y_i^2 + 8y_i - 1$$

Summing over the four roots,

$$q_4 - 4q_3 + 6q_2 - 8q_1 + 4 = 0$$

Substitute $q_1 = 4$, $q_2 = 4$, $q_3 = 16$:

$$\begin{aligned} q_4 - 4(16) + 6(4) - 8(4) + 4 &= 0 \\ q_4 - 64 + 24 - 32 + 4 &= 0 \\ q_4 - 68 &= 0 \end{aligned}$$

So

$$q_4 = 68$$

Hence

$$\alpha^8 + \beta^8 + \gamma^8 + \delta^8 = 68$$

14. The equation $z^4 - 2z^3 - z^2 - 2z + 1 = 0$ has roots α, β, γ and δ .

(a) Show that a quartic equation whose roots are $\alpha + \frac{1}{\alpha}, \beta + \frac{1}{\beta}, \gamma + \frac{1}{\gamma}$ and $\delta + \frac{1}{\delta}$ is

$$w^4 - 4w^3 - 2w^2 + 12w + 9 = 0 \quad [3]$$

(b) Hence determine the exact roots of the equation $z^4 - 2z^3 - z^2 - 2z + 1 = 0$. [3]

Solution

(a) Let

$$f(z) = z^4 - 2z^3 - z^2 - 2z + 1$$

and let z be any root of $f(z) = 0$.

Since the constant term is 1, no root is 0, so we may divide by z^2 :

$$\begin{aligned} z^2 - 2z - 1 - \frac{2}{z} + \frac{1}{z^2} &= 0 \\ \left(z^2 + \frac{1}{z^2}\right) - 2\left(z + \frac{1}{z}\right) - 1 &= 0 \end{aligned}$$

Now let

$$w = z + \frac{1}{z}$$

Then

$$z^2 + \frac{1}{z^2} = \left(z + \frac{1}{z}\right)^2 - 2 = w^2 - 2$$

so

$$\begin{aligned} (w^2 - 2) - 2w - 1 &= 0 \\ w^2 - 2w - 3 &= 0 \\ (w - 3)(w + 1) &= 0 \end{aligned}$$

So the possible values of w are 3 and -1 .

Also, the polynomial is unchanged when its coefficients are reversed, so if z is a root then $\frac{1}{z}$ is also a root. Therefore the four roots $\alpha, \beta, \gamma, \delta$ occur in reciprocal pairs, and the four values

$$\alpha + \frac{1}{\alpha}, \quad \beta + \frac{1}{\beta}, \quad \gamma + \frac{1}{\gamma}, \quad \delta + \frac{1}{\delta}$$

are

$$3, 3, -1, -1$$

Hence the required quartic is

$$\begin{aligned} (w - 3)^2(w + 1)^2 &= (w^2 - 2w - 3)^2 \\ &= w^4 - 4w^3 - 2w^2 + 12w + 9 \end{aligned}$$

So the quartic equation is

$$w^4 - 4w^3 - 2w^2 + 12w + 9 = 0$$

(b) From part (a), the corresponding values of

$$w = z + \frac{1}{z}$$

are 3 and -1 .

If $w = 3$, then

$$\begin{aligned}z + \frac{1}{z} &= 3 \\ z^2 - 3z + 1 &= 0\end{aligned}$$

So

$$z = \frac{3 \pm \sqrt{5}}{2}$$

If $w = -1$, then

$$\begin{aligned}z + \frac{1}{z} &= -1 \\ z^2 + z + 1 &= 0\end{aligned}$$

So

$$z = \frac{-1 \pm i\sqrt{3}}{2}$$

Hence the exact roots of

$$z^4 - 2z^3 - z^2 - 2z + 1 = 0$$

are

$$\frac{3 + \sqrt{5}}{2}, \quad \frac{3 - \sqrt{5}}{2}, \quad \frac{-1 + i\sqrt{3}}{2}, \quad \frac{-1 - i\sqrt{3}}{2}$$

15. The quartic equation $x^4 - 2x^3 + 3x^2 - 4x + 1 = 0$ has roots $\alpha, \beta, \gamma, \delta$.

(a) Find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta}$. [2]

(b) Find the value of $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2} + \frac{1}{\delta^2}$. [2]

(c) Find the value of

$$\left(\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}\right)^2 + \left(\frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta}\right)^2 + \left(\frac{1}{\gamma} + \frac{1}{\delta} + \frac{1}{\alpha}\right)^2 + \left(\frac{1}{\delta} + \frac{1}{\alpha} + \frac{1}{\beta}\right)^2$$
 [5]

Solution

(a) For the monic quartic

$$x^4 - 2x^3 + 3x^2 - 4x + 1 = 0$$

with roots $\alpha, \beta, \gamma, \delta$, the standard root relations give

$$\alpha + \beta + \gamma + \delta = 2$$

$$\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = 3$$

$$\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta = 4$$

$$\alpha\beta\gamma\delta = 1$$

Hence

$$\begin{aligned} \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta} &= \frac{\beta\gamma\delta + \alpha\gamma\delta + \alpha\beta\delta + \alpha\beta\gamma}{\alpha\beta\gamma\delta} \\ &= \frac{4}{1} \\ &= 4 \end{aligned}$$

(b) Let

$$r_1 = \frac{1}{\alpha}, \quad r_2 = \frac{1}{\beta}, \quad r_3 = \frac{1}{\gamma}, \quad r_4 = \frac{1}{\delta}$$

Then from part (a),

$$r_1 + r_2 + r_3 + r_4 = 4$$

Also

$$\begin{aligned} r_1r_2 + r_1r_3 + r_1r_4 + r_2r_3 + r_2r_4 + r_3r_4 &= \frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\alpha\delta} + \frac{1}{\beta\gamma} + \frac{1}{\beta\delta} + \frac{1}{\gamma\delta} \\ &= \frac{\gamma\delta + \beta\delta + \beta\gamma + \alpha\delta + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma\delta} \\ &= \frac{\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta}{\alpha\beta\gamma\delta} \\ &= \frac{3}{1} \\ &= 3 \end{aligned}$$

Now use

$$(r_1 + r_2 + r_3 + r_4)^2 = r_1^2 + r_2^2 + r_3^2 + r_4^2 + 2(r_1r_2 + r_1r_3 + r_1r_4 + r_2r_3 + r_2r_4 + r_3r_4)$$

So

$$\begin{aligned}\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2} + \frac{1}{\delta^2} &= 4^2 - 2(3) \\ &= 16 - 6 \\ &= 10\end{aligned}$$

(c) Using the same notation, let

$$S = r_1 + r_2 + r_3 + r_4 = 4$$

Then each bracket is the sum of all four reciprocals except one:

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = S - r_4$$

$$\frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta} = S - r_1$$

$$\frac{1}{\gamma} + \frac{1}{\delta} + \frac{1}{\alpha} = S - r_2$$

$$\frac{1}{\delta} + \frac{1}{\alpha} + \frac{1}{\beta} = S - r_3$$

So the required expression is

$$(S - r_1)^2 + (S - r_2)^2 + (S - r_3)^2 + (S - r_4)^2$$

Expanding,

$$\begin{aligned}\sum_{i=1}^4 (S - r_i)^2 &= \sum_{i=1}^4 (S^2 - 2Sr_i + r_i^2) \\ &= 4S^2 - 2S \sum_{i=1}^4 r_i + \sum_{i=1}^4 r_i^2\end{aligned}$$

From parts (a) and (b),

$$\sum_{i=1}^4 r_i = S = 4 \quad \text{and} \quad \sum_{i=1}^4 r_i^2 = 10$$

Therefore

$$\begin{aligned}\sum_{i=1}^4 (S - r_i)^2 &= 4(4^2) - 2(4)(4) + 10 \\ &= 64 - 32 + 10 \\ &= 42\end{aligned}$$

So the value of the expression is

$$42$$